Exercise 1: Computing $\cos x$ for $0 \le x \le 2\pi$

Trigonometric functions are periodic functions. The graphs of these functions repeat after a certain time interval, called time period, and generally represented by T. The time period of $\cos x$ is 2π.

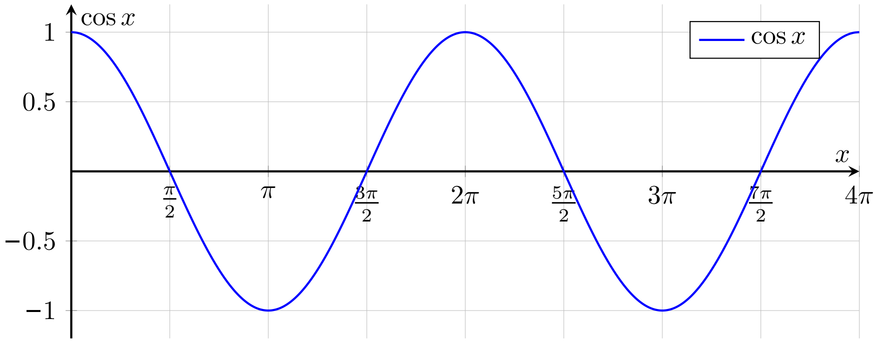

Figure 1: Graph of $\cos x$.

Mathematically $\cos(x + 2n\pi) = \cos x$, where n is any positive or negative integer. The series expansion of $\cos x$ is

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + - \cdots = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k}}{(2k)!} \tag{1}$$

$$= 1 + 1 \times \left(-\frac{x^2}{2 \times 1}\right) + 1 \times \left(-\frac{x^2}{2 \times 1}\right) \times \left(-\frac{x^2}{4 \times 3}\right) + \cdots \tag{2}$$

In this exercise you will write a function to compute $\cos x$ for any positive or negative value of x. Since $\cos x$ is a periodic function, you will first scale x so that it is between 0 and 2π. As an example suppose $x = 10.2$rad. This value is greater than 2π. Hence, we subtract 2π from this value to get $10.1 - 2\pi \approx 3.8168$. The value of $\cos 10.1 = \cos(10.1 - 2\pi) \approx \cos 3.8168$. Instead of calculating $\cos 10.1$, we calculate $\cos 3.8168$. If, for example, $x = 29.9$, we subtract 2π to get 23.6168. This value is still greater than 2π so we subtract 2π from this value again, to get 17.3336. We continue subtracting 2π till we get a value between 0 and 2π. For the negative values of x we repeatedly add 2π to the value x till we get a value between 0 and 2π.

1. Receive double precision x from keyboard
2. If x is not between 0 and 2π, repeatedly add or subtract 2π from x till it is between 0 and 2π
3. Use this value of x to calculate $\cos x$ by using the series expansion of Equation 2.
4. Return the value to caller

The value of $\cos x$ must be accurate to 6 decimal places. You know how to do this. Right? Use a constant `delta` with a value of 0.0000001. Please strictly adhere to the MIPS function calling and register usage conventions. Name of the function must be `cos`.

Floating point parameters

If the floating point parameters to be passed to a function are single-precision, then the first one is passed in register `$f12` and the second one is passed in register `$f14`. The first double-precision parameter is passed in register `$f12` and `$f13`, and the second double-precision parameter is passed in register `$f14` and `$f15`. If more than two floating point parameters are to be passed to a function, then rest are passed on stack. We may talk about passing parameters on stack later in the course.

Submission Instructions

Submit the following file.

1. `<rollnumber>-h3-ex1.asm` containing solution of exercise 1

Strictly following the above naming convention. You will certainly lose marks otherwise.